

Research Gaps for an AAAI Knowledge-Graph Paper: An Assessment of Rishii Kumar Singh's Work Against the AAAI 2025/2026 KG Landscape

TL;DR

- The single strongest, most defensible AAAI-caliber gap is the one Rishii has already partly evidenced: **"self-healing knowledge bases" — detecting and repairing AI-fabricated entity descriptions inside a multilingual KB using the graph's own structure, concentrated in low-resource/non-Latin scripts, with a measured downstream effect on completion/grounding**. An independent literature check plus my own confirm this exact four-part combination is unoccupied at AAAI 2025/2026 and adjacent top venues.
- Rishii is unusually well-positioned: he already has (a) a working multilingual inductive KGC "substrate" (BGE-M3 + LoRA + CRR loss, 26.51 ± 0.31 MRR over 3 seeds on his DBP-5L-Ind benchmark), (b) a hand-audited finding that ~70% of his LLM-backfilled descriptions were fabricated, (c) a training-free structure-vs-text detector at ROC-AUC 0.995, and (d) a repair result (+1.95 MRR on Japanese). ~80% of the needed experiments already exist.
- His own docs already target WSDM 2027 (full paper ~Aug 24, 2026), not AAAI-27 (full paper Jul 28, 2026), because the AAAI window is too tight for a solo author. My recommendation: keep the self-healing paper on the WSDM/KDD track and treat the finished substrate as a **separate, lower-risk AAAI-27 benchmark paper** if he wants an AAAI submission specifically.

Key Findings

1. What Rishii's research actually is. His "Knowledge Graph" project has two connected parts. The *substrate* (built and evaluated) is multilingual **inductive** knowledge-graph completion: a BGE-M3 text bi-encoder fine-tuned with LoRA (rank 16), CRR ranking loss, and hard negatives, completing facts about unseen entities across five languages (English, French, Spanish, Japanese, Greek — Greek low-resource) on a single 16 GB RTX 5070 Ti. He built the first entity-disjoint inductive split of DBP-5L ("DBP-5L-Ind": 35,848 train / 8,959 unseen-test entities). Headline result: 26.51 ± 0.31 MRR over 3 seeds (18× over zero-shot's 1.45; beats an mBERT rerun at 24.08). The *pivot* (the new headline) is **self-healing knowledge bases**: he discovered his own KB was ~70% contaminated with confident LLM-generated fabrications (Llama 3.2 back-filling descriptions for entities lacking Wikipedia coverage — e.g., "Clint Eastwood" in Greek became "a Greek footballer for PAOK"; "Taipei" became "a Cypriot dessert"), concentrated in low-resource Greek/Japanese, and that the graph's own structure detects and repairs it.

2. His documented open questions and limitations. From his repo docs: the substrate's contribution is "a benchmark plus first baselines plus empirical findings, not a new architecture" (his senior's central critique). Structural auditing only works for entities that *have* graph edges —

brand-new entities with a description but no edges cannot be audited. The method assumes structure is more trustworthy than text (true in curated KGs, stated as an assumption). Generality needs a second KB (a Wikidata slice) — flagged as future work. The detector currently flags only 57% of *real* fabrications because those low-resource entities have almost no structure to check against — the honest empirical soft spot.

3. The AAAI 2025/2026 KG landscape is dominated by nine clusters, and Rishii's gap sits in the smallest one. The dominant themes: KG completion/embeddings; inductive KGC; temporal KGs; multimodal KGs; LLM+KG and GraphRAG (the fastest-growing cluster); KGQA/reasoning; entity alignment; hyper-relational/n-ary KGs; and the smallest — **KG error detection/denoising/quality**. Representative named works: KG CRR (AAAI 2025, the CRR loss Rishii uses); Relation-Aware Anchor KGC (AAAI 2025); TruthfulRAG (AAAI 2026); "You Don't Need Pre-built Graphs for RAG" (AAAI 2026); KG Error Detection with Contrastive Confidence Adaption (AAAI 2024); Type-Information-Assisted Self-Supervised KG Denoising (AISTATS 2025). For scale: AAAI 2026 accepted exactly 4,167 of 23,680 submissions (17.6% acceptance rate, the lowest in three years, up from 12,957 submissions / 23.4% at AAAI 2025). The competitive center of gravity is LLM+KG/GraphRAG — crowded and risky. KG quality/denoising is thin and is exactly where Rishii's evidence lives. (AAAI + 5)

4. Novelty verdict (independently confirmed): the self-healing gap is open. No AAAI 2025/2026 paper, and no verified SIGIR/KDD/WSDM/WWW/ACL/EMNLP 2025–2026 paper, combines all four pillars: (1) detecting AI-fabricated contamination in a KB's *own* entity descriptions, (2) using the KG's own structure (structure-vs-text disagreement) as the signal, (3) concentrated in low-resource/non-Latin scripts, (4) with a measured downstream completion/grounding effect plus repair. The nearest prior art each misses ≥ 2 pillars (see Details).

Details

The current state of the art, mapped to Rishii's assets

The four closest papers and their precise gaps:

- **KG Error Detection with Contrastive Confidence Adaption (CCA)** — Xiangyu Liu, Yang Liu & Wei Hu (Nanjing University), AAAI 2024, Vol. 38(8), pp. 8824–8831 (doi:10.1609/aaai.v38i8.28729). Closest on mechanism: "CCA to integrate both textual and graph structural information from triplet reconstruction ... we construct realistic datasets with semantically-similar noise and adversarial noise" — confirming its noise is synthetic, not AI-fabricated. It is also monolingual English, detection-only (no repair), and measures no downstream impact. This is the paper reviewers will most cite against him; his differentiation ("finds bad *edges*; we find AI-fabricated *descriptions*") must be explicit.
- **Type-Information-Assisted Self-Supervised KG Denoising (AISTATS 2025, arXiv 2503.09916)** — structure-based self-supervised noise detection, but uses *no* text descriptions (so no structure-vs-text disagreement), no multilingual angle, no repair, no downstream grounding. (arXiv)

- **TruthfulRAG (AAAI 2026)** — uses KGs to resolve factual conflicts, but at RAG *inference* time between sources, not contamination baked into a stored KB; no low-resource focus.
- **Retrieval Collapses When AI Pollutes the Web (WWW 2026, short paper)** — proves AI pollution degrades retrieval (67% pool contamination → 80%+ exposure contamination; accuracy stays deceptively stable while source diversity collapses) but proposes no detection, no KG signal, no repair. It *motivates* Rishii's direction rather than occupying it.

arXiv

Adjacent but further:

- **mRAKL** — Hellina Hailu Nigatu, Min Li, Maartje Ter Hoeve, Saloni Potdar & Sarah Chasins (UC Berkeley + Apple), Findings of ACL 2025 (pp. 13072–13089). "With an idealized retrieval system, mRAKL improves accuracy by up to 4.92 and 8.79 percentage points for Tigrinya and Amharic, respectively"; it contributes Tigrinya (3.5k triples) and Amharic (34k triples) KGs. It *constructs* low-resource multilingual KGs via RAG and assumes evidence is trustworthy — the opposite of Rishii's threat model.
- **KG-TRICK** — Zelin Zhou, Simone Conia et al. (Apple), COLING 2025 (2025.coling-main.611), an mBART-large-50 seq-to-seq framework introducing "WikiKGE10++, the largest manually-curated benchmark for textual information completion of KGs, which features over 25,000 entities across 10 diverse languages." It *fills missing* text/links; it does not audit contamination.
- **FactNet (2026)** — a billion-scale multilingual grounding resource that *prevents* contamination by deterministic construction rather than detecting/repairing it.

Hugging Face

Hugging Face

The prioritized gaps

GAP 1 (TOP PRIORITY — novel, defensible, evidence-in-hand): Self-healing multilingual knowledge bases.

- *What it is:* Name and formalize the phenomenon of structural self-contamination of multilingual KBs by generated text; build a training-free structural detector (serialize an entity's graph neighbourhood, score description consistency via NLI and LLM-judge variants); quarantine + re-source; measure downstream completion/grounding gains.
- *Evidence it's a gap:* Section 4 and the four-paper analysis above; independently confirmed white space.
- *Why it matters:* Anyone training on or building RAG over scraped multilingual KBs is ingesting confident AI falsehoods now, worst in low-resource languages, with no working defence.
- *How his work positions him:* He owns the audit (~70% fabricated), the detector (ROC-AUC 0.995, F1 0.988 on a controlled benchmark), the repair (+1.95 MRR Japanese), the

benchmark (DBP-5L-Ind), and the retriever (the substrate model). ~80% of experiments exist.

- *Proposed method/eval*: Characterize contamination by language/script; detect (NLI + judge variants, precision/recall by language); repair (re-source from native Wikipedia, which recovered 45% for him); downstream LLM factual-QA with vs. without cleaned KB. Add a *realistic-fabrication* injection set (believable wrong bios, not wrong-entity swaps) and a second KB (Wikidata slice) for generality.
- *Feasibility*: Tight but real for WSDM 2027 (~6-7 weeks from his July 2026 state). Too tight for AAAI-27's Jul 28, 2026 full-paper deadline as the lead paper.
- *Risk*: Medium. Must differentiate crisply from CCA (AAAI 2024) and the AISTATS 2025 denoiser. The "no structure to check" limit (only 57% of real fabrications flagged) is the main empirical soft spot.

GAP 2 (strong, lower-risk, AAAI-fit): A multilingual *inductive* KGC benchmark + first baselines.

- *What it is*: Release DBP-5L-Ind — the first entity-disjoint inductive split of a multilingual KG — with the first strong inductive baselines and the finding that all prior transductive multilingual methods (KEnS, AlignKGC, SS-AGA) structurally cannot run on unseen entities.
- *Evidence it's a gap*: The intersection of inductive + multilingual + low-resource was verified open through EMNLP 2025 in his own literature sweep; the transductive-only limitation of prior multilingual KGC is real.
- *Why it matters/positions him*: This is fully built, seed-stable, significance-tested (+12.67 MRR vs pre-enrichment, $p < 0.001$), with cross-lingual robustness and transductive-ceiling comparisons. It maps cleanly to AAAI's "DMKM: Datasets & Benchmarks" track.
- *Feasibility*: Highest — essentially done. Best candidate if he specifically wants an AAAI-27 submission.
- *Risk*: The senior's critique ("architecture not novel; dataset swap") applies — must be framed benchmark-first, not SOTA-first.

GAP 3 (extension, higher-novelty-ceiling): Auditing contamination for structure-poor entities.

- His detector's weakness (can't audit entities with no edges — the exact low-resource case) is itself a research gap: cross-lingual structural transfer (borrow aligned high-resource neighbourhoods) or hybrid stylometric+structural signals to audit edge-sparse entities. Novel, but less mature — a follow-up, not the first paper.

GAP 4 (crowded/risky — note as such): Completion-before-RAG for multilingual factuality.

- His own "Direction B." mRAKL and KG-TRICK already occupy most of this; incremental over published work. He correctly de-prioritized it. Flagged here so he doesn't revisit it as a

headline.

Venue timing (verified)

- **AAAI-27:** abstracts due July 21, 2026; full papers due July 28, 2026; conference February 16–23, 2027, Montréal, Canada. Submissions limited to 7 pages of main content, max 9 pages total; no CJK package (Greek/Japanese examples need romanization or figure-embedded text).
- **WSDM 2027:** Cordis, Hong Kong, February 15–19, 2027 (20th ACM WSDM); his docs cite full-paper ~Aug 24, 2026, with a Findings safety net. This is his stated target for the pivot.
- **KDD 2027 Cycle 2** (~Feb 2027) and **SIGIR 2027** are fallbacks; both fit KG data-quality work.

Recommendations

1. **Lead with Gap 1 (self-healing) for WSDM 2027**, exactly as his docs plan. Benchmark that would change this: if a competing AAAI-26/WWW-26/SIGIR-26 paper surfaces that couples structural contamination *detection + repair* on multilingual KBs, pivot the framing toward Gap 3 (structure-poor auditing), where the field is emptier.
2. **If an AAAI-27 submission is specifically desired, submit Gap 2 (DBP-5L-Ind benchmark) to AAAI-27's Datasets & Benchmarks track** — it is done, defensible, and fits the 7-page format. Keep Gap 1 for WSDM to avoid splitting the novel result across a too-tight AAAI deadline.
3. **Harden Gap 1 before submission:** (a) build the realistic-fabrication injection set; (b) add the LLM-judge detector variant beside the embedding one; (c) run the downstream factual-QA experiment; (d) add a Wikidata slice for generality. Threshold: if detector recall on structure-poor entities stays near 57%, foreground the honest coverage limit and lead with the downstream-grounding result as the headline instead of raw detection.
4. **Differentiate explicitly** from CCA (AAAI 2024) and the AISTATS 2025 denoiser in the related-work section; reviewers will anchor there.
5. **Verify every arXiv ID and venue** before submission — his own docs flag that some mid-2026 search-surfaced IDs may be imprecise.

Caveats

- I could not access a `project_knowledge_search` tool in this environment; my reconstruction of the research journal relies on Rishii's GitHub repo (README, PROJECT_STORY_AND_PLAN.md, IMPLEMENTATION_PLAN.md), which are detailed and internally consistent. The 37 AAAI PDFs themselves were not directly readable; the theme map is built from public AAAI proceedings/aggregators, so specific per-PDF content may differ from what is in his project folder.

- AAI 2026 accepted 4,167 papers; neither I nor the verification subagent could enumerate all of them, so a niche partial-overlap KG-quality paper could exist. Confidence in the "open" verdict for the exact four-pillar combination is high; medium that zero partially overlapping AAI-26 papers exist.
- Several numbers (70% contamination, ROC-AUC 0.995, +1.95 MRR, 26.51 MRR) are Rishii's own reported results from his repo, not independently reproduced. His docs note some are single-run or on controlled benchmarks.
- His repo is dated "as of 2026-07-07," so the competitive landscape may have shifted; a fresh literature sweep immediately before submission is essential.